UNA UNIVERSIDAD NACIONAL COSTA RICA

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Advances in Artificial Intelligence – IBERAMIA 2026: 19th Ibero-American Conference on Artificial Intelligence, Asuncion, Paraguay, November 18–20, 2026, Proceedings– Accept Email:A Comparative Evaluation of Bio-Inspired Hyperparameter Optimization Algorithms for Software Defect Density Prediction

1 mensaje

Sent on behalf of José Manuel Ferreira Machado 22 de julio de 2026 a las 9:01
<meteor.info@springer.com>
Responder a: José Manuel Ferreira Machado <jmac@di.uminho.pt>
Para: alessandro.cambronero.delgado@est.una.ac.cr

Dear Alessandro Cambronero Delgado,
Thank you again for your submission to the project Advances in Artificial Intelligence – IBERAMIA 2026: 19th Ibero-American Conference on Artificial Intelligence, Asuncion, Paraguay, November 18–20, 2026, Proceedings. It's my pleasure to accept your manuscript.

Review 1 (Acceptance)
The paper presents a solid empirical study evaluating four bio-inspired optimizers (GA, PSO, ABC, CDEA) across 11 regression models using PROMISE datasets. The methodology follows the GQM paradigm and the statistical analysis using the Friedman test is rigorous and transparent.

The English is correct, and there are no spelling or formatting errors.

I recommend accepting, considering minor revisions.
My suggestion is listed below:

- Optimizers' hyperparameters: please detail the internal hyperparameters of the bio-inspired optimizers themselves (e.g., population size, mutation rates, stopping criteria) to ensure the execution time comparison is fully benchmarked under identical search efforts.

- Complementary metrics: consider adding MAE to the discussion to better assess how the models handle outliers or highly skewed distributions in software defect density.

- Search space disparity: the number of tuned hyperparameters varies significantly across the 11 regression models. Please discuss how the dimensionality of each model's search space affects the optimizers' execution times and convergence behavior.

- Dataset modernization: since the PROMISE repository consists primarily of legacy systems, please expand the discussion on external validity regarding how these bio-inspired tuning mechanisms would scale or adapt to modern software development contexts (e.g., continuous deployment or microservices).

Review 2 (Acceptance)
The manuscript is generally clear, well structured, and relevant to the topics of IBERAMIA. However, several aspects of the experimental design, statistical analysis, and reproducibility require further clarification.

The description of the experimental design should be more explicit. The manuscript refers to ten datasets, eleven regression algorithms, four optimization algorithms, and ten random data partitions, while reporting a total of 550 execution records. It would be desirable to explain in detail how this total was obtained, what exactly constitutes an execution record, how many executions were conducted for each optimizer, and what experimental unit was used in the statistical analyses. A clearer breakdown of the experimental matrix would facilitate the readers' understanding and assessment of the results.

Additional information about the configuration of the optimization algorithms would enhance the reproducibility of the study. Specifically, the manuscript should report the population or swarm sizes, number of iterations, evaluation budgets, hyperparameter search ranges, encoding of categorical and discrete variables, stopping criteria, random seeds, and relevant implementation details. This information is also necessary to determine whether the execution-time comparison was conducted under equivalent computational conditions.

The interpretation of the statistical results also requires clarification. The manuscript describes the predictive performance of the optimizers as “statistically equivalent” based on a non-significant Friedman test. It would be more precise to state that “no statistically significant differences were found,” unless a formal equivalence test is performed. Additionally, since the Friedman test for execution time was statistically significant, a post hoc analysis would be advisable to identify which specific pairs of algorithms differ.

The manuscript would also benefit from a more explicit explanation of its contribution relative to the previously published framework. As the study extends earlier work by incorporating additional optimization algorithms, the authors should clarify which aspects of the experimental design and results are novel.

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A Comparative Evaluation of Bio-Inspired Hyperparameter Optimization Algorithms for Software Defect Density Prediction
We kindly request that during the final submission, please ensure the source files (LaTeX/Docx), PDF, and figures are also uploaded. Please take a moment to delete any irrelevant files, such as outdated versions that are not part of this submission. These source files are required for preparing the proofs during the production process by the typesetters.

Additionally, we ask you to check the names, affiliations, and the order of the authors is correct in the paper and on the Meteor system.

With thanks and best regards,
Conference Organizing Committee